Practical-3

Problem Definition:

Adigital thermostatreceives two 4-bit temperature readings:

- T_room (temperature inside the room)
- T_safe (maximum safe temperature threshold)

Design a digital system that:
1. Compares T_room and T_safe.
2. Activates an overheat alarm (A = 1) if T_room > T_safe.
3. Activates a ventilation system (V = 1) if T_room ≥ T_safe – 2 degrees.
4. Triggers a system shutdown (S = 1) if T_room ≥ T_safe + 3 degrees.
5. Ensures that if multiple outputs activate, priority is: Shutdown > Alarm > Ventilation

Tools/Technology To Be Used:

- Circuit Design:Logisim,Multisim,TinkerCad (if logic components available), or Verilog
- Logic Simplification: K-map templates, Boolean Algebra rules

Aim:

To design and implement a digital thermostat system that compares two 4-bit temperature inputs (T_room and T_safe) and generates appropriate control signals—Overheat Alarm (A), Ventilation (V), and System Shutdown (S)—based on defined temperature conditions, using digital logic design techniques and simulation tools such as Logisim or Verilog.

2-Bit Magnitude Comparator

Description:

A **2-bit magnitude comparator** is a combinational logic circuit used to compare two 2-bit binary numbers and determine their relative magnitude. It takes two inputs, **A (A1 A0)** and **B (B1 B0)**, and produces three outputs indicating whether one number is greater than, equal to, or less than the other.

The comparator generates:

G (A > B) when the value of A is greater than B

E (A = B) when both numbers are equal

L (A < B) when A is less than B

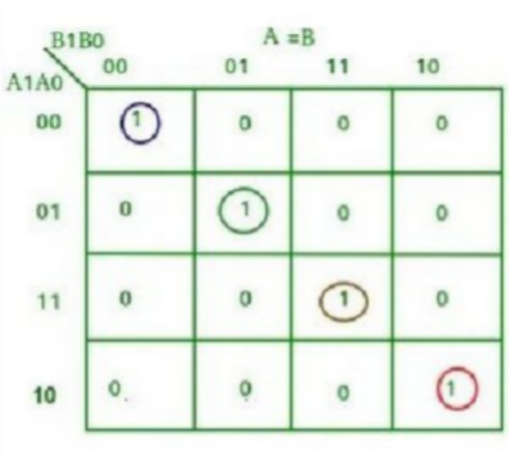
The comparison is performed by first checking the ***most significant bits (MSB)***, A1 and B1. If they are different, the result is decided immediately based on these bits. If the MSBs are equal, then the ***least significant bits (LSB)***, A0 and B0, are compared to determine the final result.
This approach makes the circuit efficient, as it avoids unnecessary comparisons once a decision is made at the higher bit level. The comparator is implemented using basic logic gates such as AND, OR, NOT, and XNOR.

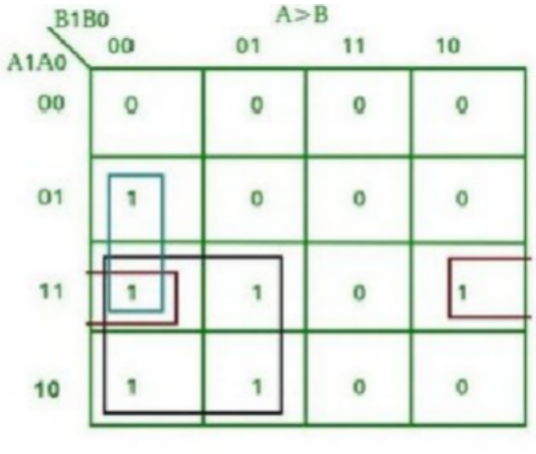
2-bit magnitude comparators are commonly used in digital systems such as arithmetic logic units (ALUs), decision-making circuits, and data comparison operations in processors.

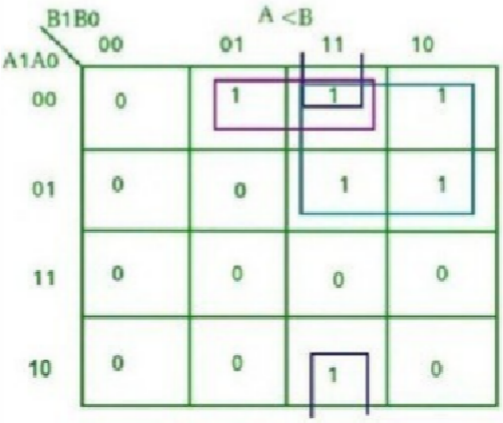
Truth table:

A1	A0	B1	B0	G	E	L
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

K-Map:







Equation:

GreaterThanEquation : $A_1B_1' + A_0B_1'B_0' + A_1A_0B_0'$

EqualityEquation: $A_1'A_0'B_1'B_0' + A_1'A_0B_1'B_0 + A_1A_0B_1B_0 + A_1A_0'B_1B_0'$

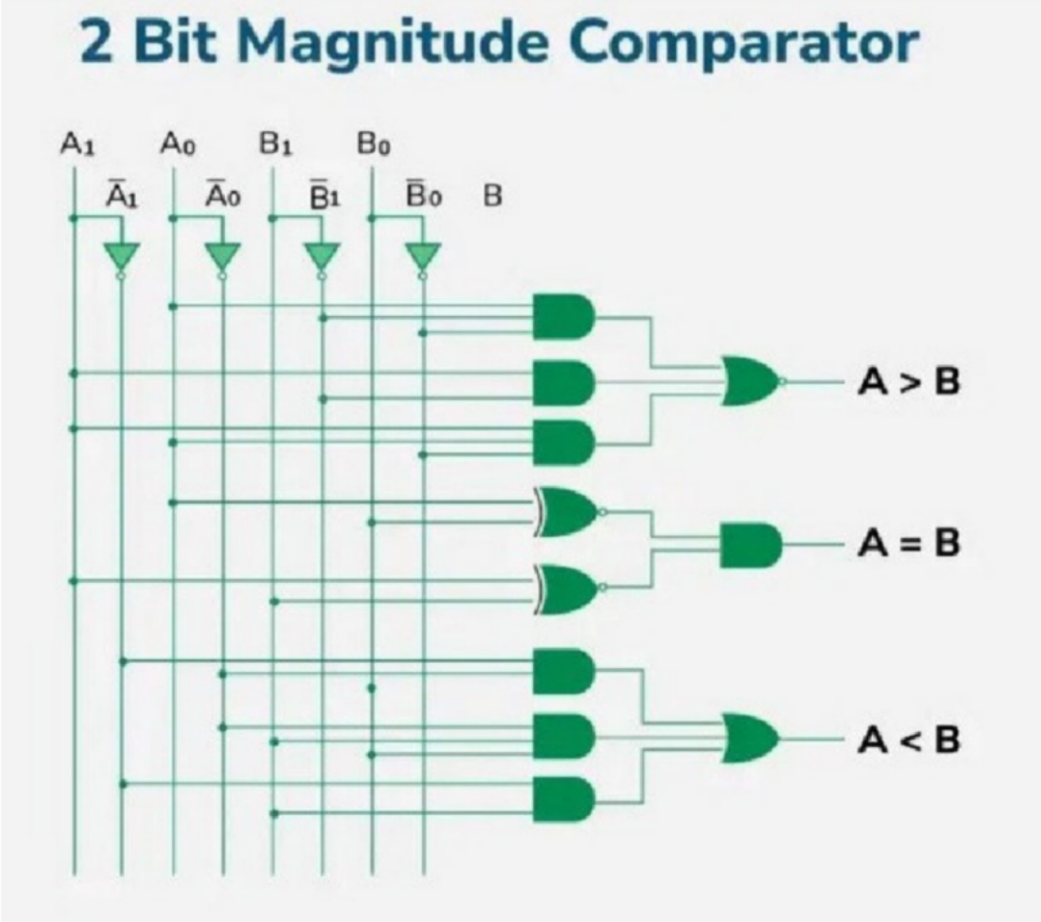
$:A_1'B_1' (A_0'B_0' + A_0B_0) + A_1B_1 (A_0B_0 + A_0'B_0')$

$:(A_0B_0 + A_0'B_0') (A_1B_1 + A_1'B_1')$

$:(A_0$ Ex-Nor $B_0) (A_1$ Ex-Nor $B_1)$

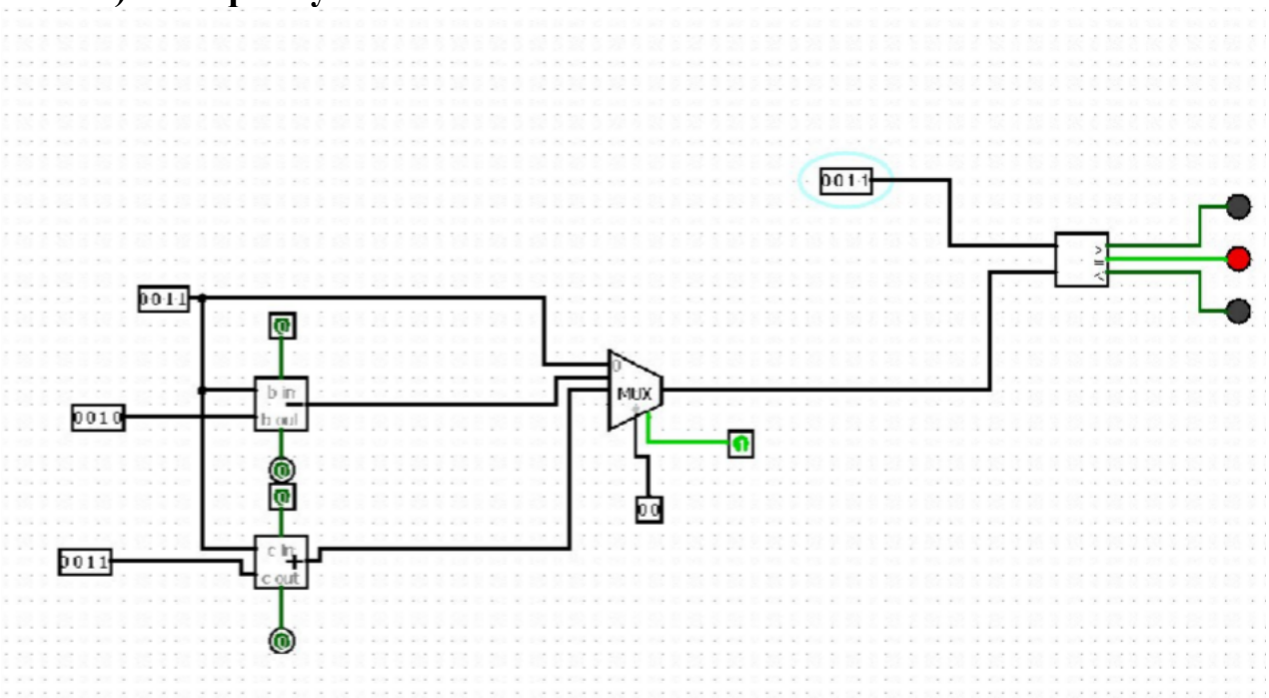
LessThanEquation: $A_1'B_1 + A_0'B_1B_0 + A_1'A0'B0$

Cricuit:

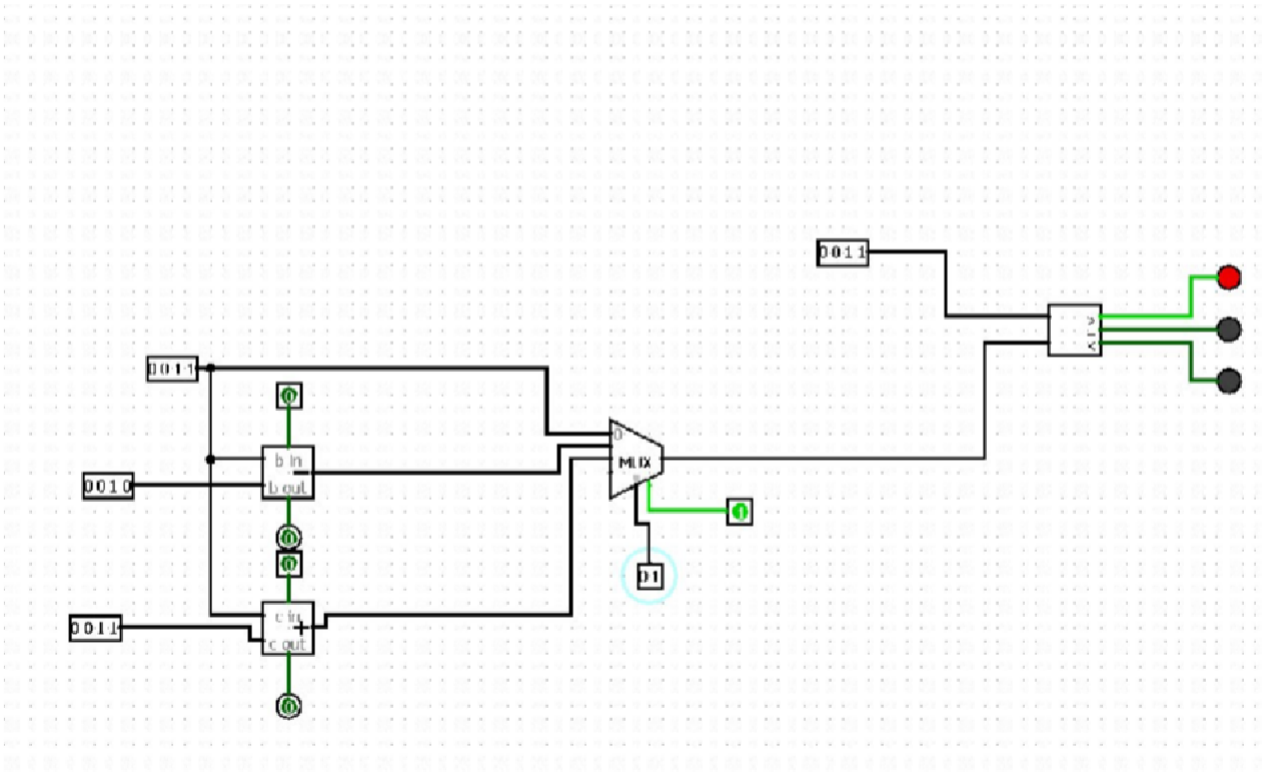


Screen shot:

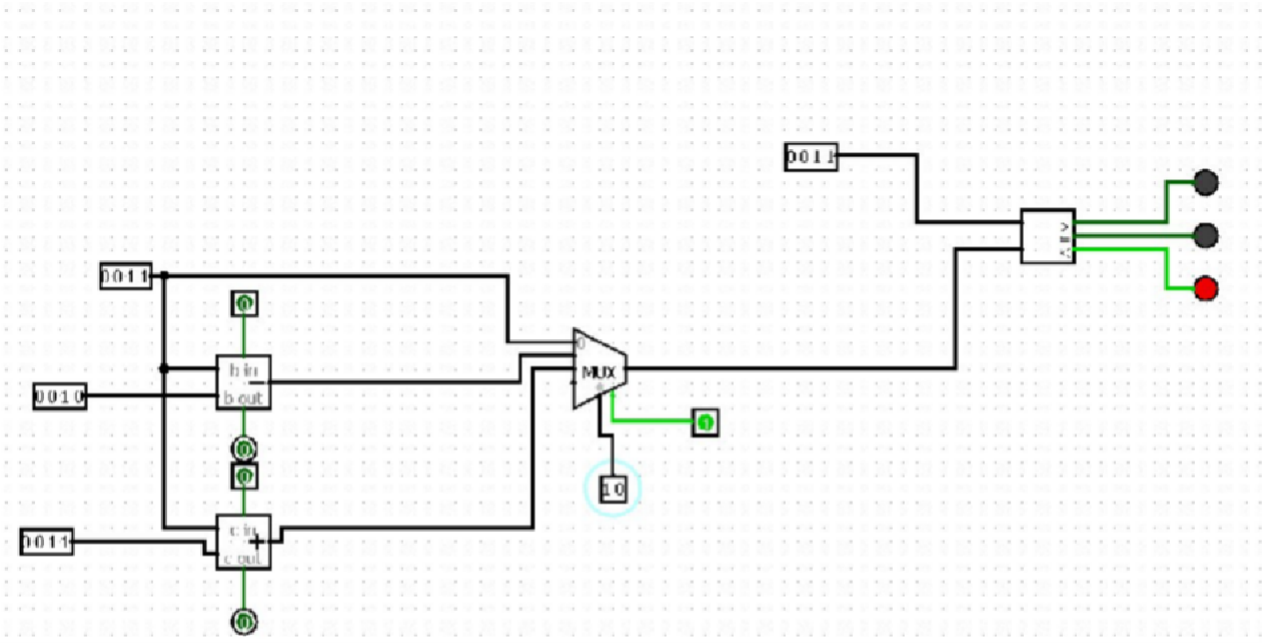
1) For equality:



2) For Greater than:



3)For Less than:



Conclusion: This digital thermostat practical demonstrates the integration of magnitude comparators and combinatorial logic to create a responsive safety system. By implementing specific arithmetic offsets (T_{safe} - 2 and T_{safe} + 3), the design proves how binary adders and subtractors can establish dynamic thresholds for environmental control. The application of priority logic ensures that critical safety states (Shutdown) override lower-level responses (Ventilation), mimicking real-world industrial fail-safes. Ultimately, the project validates the use of gate-level digital electronics in transforming raw sensor data into a structured, hierarchical decision-making system.